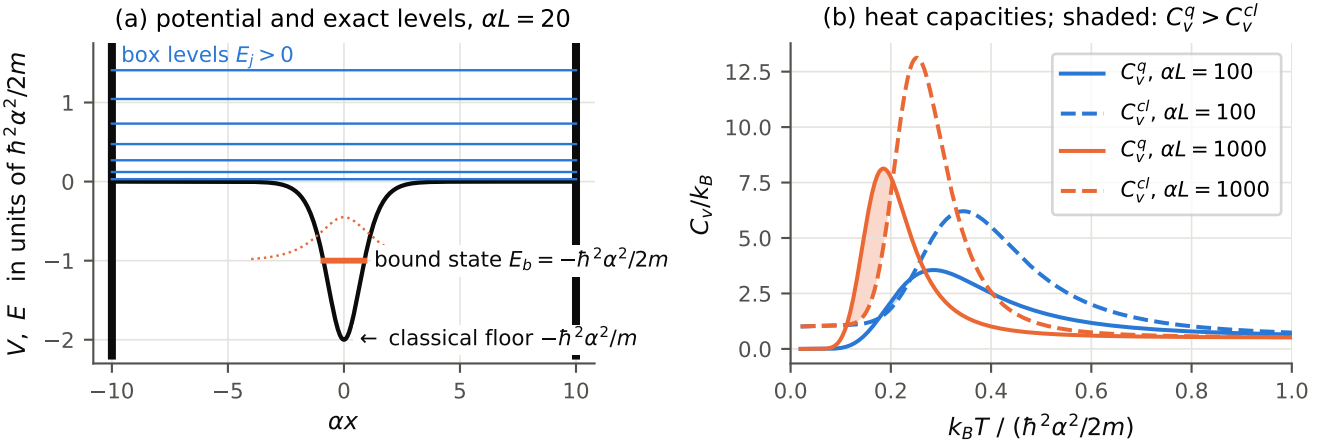
(a) potential and exact levels, $\alpha L = 20$
box levels $E_j > 0$
bound state $E_b = -\hbar^2\alpha^2/2m$
← classical floor $-\hbar^2\alpha^2/m$
$V,\ E$ in units of $\hbar^2\alpha^2/2m$
αx
(b) heat capacities; shaded: $C_v^g > C_v^{cl}$
$C_v^g,\ \alpha L = 100$
$C_v^{cl},\ \alpha L = 100$
$C_v^g,\ \alpha L = 1000$
$C_v^{cl},\ \alpha L = 1000$
C_v/k_B
$k_BT\ /\ (\hbar^2\alpha^2/2m)$